CS471 – Web Technologies (Laboratory)	Qassim University College of Computer	Lab 4
		HTML and CSS (Part 1)

This lab session covers the basics of HTML, CSS, and Django template tags. By the end of this session, students should have a very good understanding of how to build a basic website that has several html files with static files and a simple menu.

Pre-lab Preparation:

1. Having bookmodule and usermodule in apps directory, each containing separate URL configurations.
2. Template directory located in 'apps' with necessary settings.

Lab Activities:

Task 1: Build your first HTML file with Fundamental HTML Document and HTML tags (Lecture 4 – part1)

Step 1: from any location, for example, in the home directory, create a folder "CS471_lab":

```
> mkdire ~/CS471_lab
```

Step 2: Create four html files (index.html, aboutus.html, list_books.html, and one_book.html) within *CS471_lab* folder, and add the following content:

index.html

C: > Users > re2na > OneDrive > سطح المكتب > T_W > CS471_lab > index.html > html > head > title

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <title>My First Index HTML Page</title>
  <meta name="keywords" content="Books, recent, Learning, library">
  <meta name="description" content="Books website">
  <meta name="author" content="author name">
  <link rel="icon" href="qu-icon.png">
  <meta charset="UTF-8">
</head>
<body>
    <div>
      <a href="index.html">Home</a>
      <a href="list_books.html">List Books</a>
      <a href="aboutus.html">About</a>
    </div>
    <h1>This is heading 1 (Library)</h1>
    <p>This is a paragraph text </p>
    <a href="https://www.w3schools.com/html/default.asp">Learn more about HTML</a>
    <hr>
    <!-- This is a comment in HTML, not render in the bowser -->
    <h2>This is heading 2</h2>
    <p>This is some other text. <br> with newline <br> and another line</p>
    <hr>

    <h2>This is heading 2</h2>
    <p>This is some other text.</p>
</body>
</html>
```

aboutus.html

C: > Users > re2na > OneDrive > سطح المكتب > T_W > CS471_lab > aboutus.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <title>About Us</title>
  <meta name="keywords" content="Books, recent, Learning, library">
  <meta name="description" content="Books website">
  <meta name="author" content="author name">
  <link rel="icon" href="qu-icon.png">
  <meta charset="UTF-8">
</head>
<body>

    <div>
        <a href="index.html">Home</a>
        <a href="list_books.html">List Books</a>
        <a href="aboutus.html">About</a>
    </div>

    <h1>About us</h1>
    <p>This is a paragraph text This is a paragraph text This is a paragraph text This is a paragraph text </p>

    <p>To write to  <a href="mailto:email@domain.com">our store</a>, click the link and your default email client will open an email message and
</body>
</html>
```

File C:/Users/re2na/OneDrive/سطح%20المكتب/T_W/CS471_lab/aboutus.html

Home List Books About

About us

This is a paragraph text This is a paragraph text This is a paragraph text This is a paragraph text

To write to our store, click the link and your default email client will open an email message and address it to us.

list_books.html

C: > Users > re2na > OneDrive > سطح المكتب > T_W > CS471_lab > list_books.html > ...

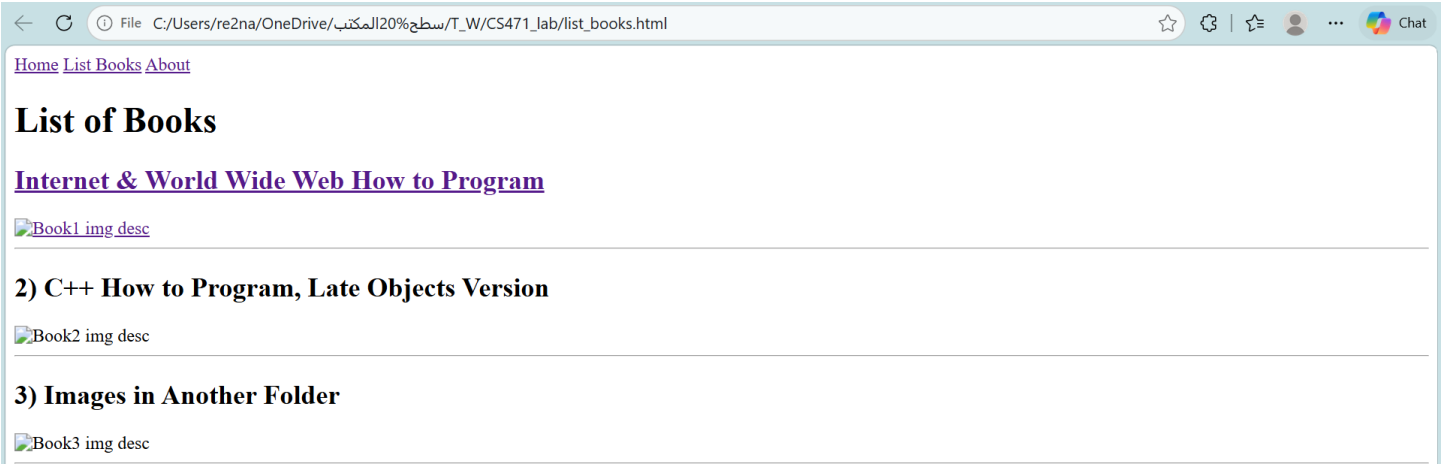
```html
<!DOCTYPE html>
<html lang="en">
<head>
  <title>List of Books</title>
  <meta name="keywords" content="Books, recent, Learning, library">
  <meta name="description" content="Books website">
  <meta name="author" content="author name">
  <link rel="icon" href="qu-icon.png">
  <meta charset="UTF-8">
</head>
<body>

    <div>
        <a href="index.html">Home</a>
        <a href="list_books.html">List Books</a>
        <a href="aboutus.html">About</a>
    </div>


    <h1>List of Books</h1>
    <a href="one_book.html"><h2>Internet & World Wide Web How to Program</h2></a>
    <a href="one_book.html"><img src="book1.jpg" alt="Book1 img desc" width="200" height="200"></a>
    <hr>

    <h2>2) C++ How to Program, Late Objects Version</h2>
    <img src="book2.jpg" alt="Book2 img desc" width="200" height="200">
    <hr>

    <h2>3) Images in Another Folder</h2>
    <img src="book3.jpg" alt="Book3 img desc" width="200" height="200">
    <hr>
```



one_book.html

C: > Users > re2na > OneDrive > سطح المكتب > T_W > CS471_lab > one_book.html > ...

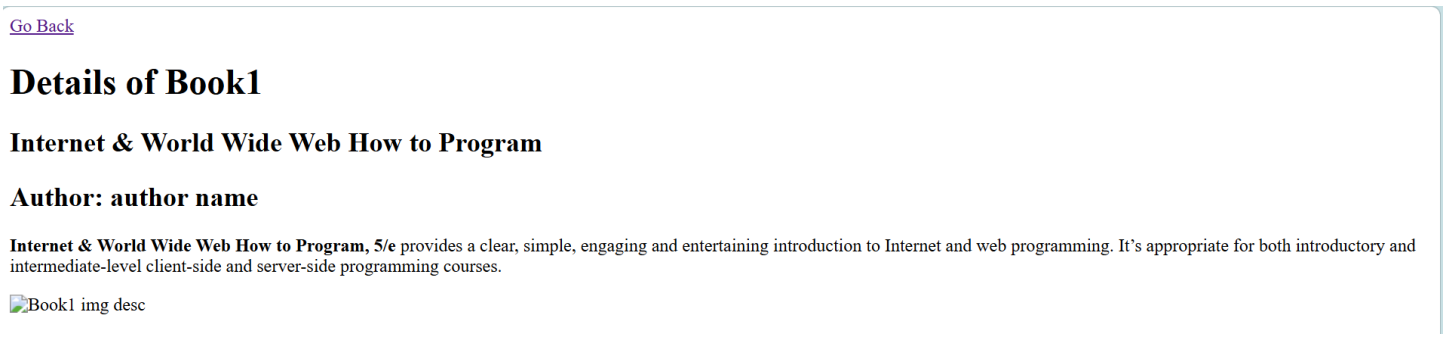
```html
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Show one Book</title>
  <meta name="keywords" content="Books, recent, Learning, library">
  <meta name="description" content="Books website">
  <meta name="author" content="author name">
  <link rel="icon" href="qu-icon.png">
  <meta charset="UTF-8">
</head>
<body>

<a href="list_books.html">Go Back</a>

<h1>Details of Book1</h1>
<h2>Internet & World Wide Web How to Program</h2>
<h2>Author: author name</h2>
<p><strong> Internet & World Wide Web How to Program, 5/e </strong> provides a clear, simple, engaging and entertaining introduction to Internet

<img src="book1.jpg" alt="Book1 img desc" width="200" height="200">

</body>
</html>
```



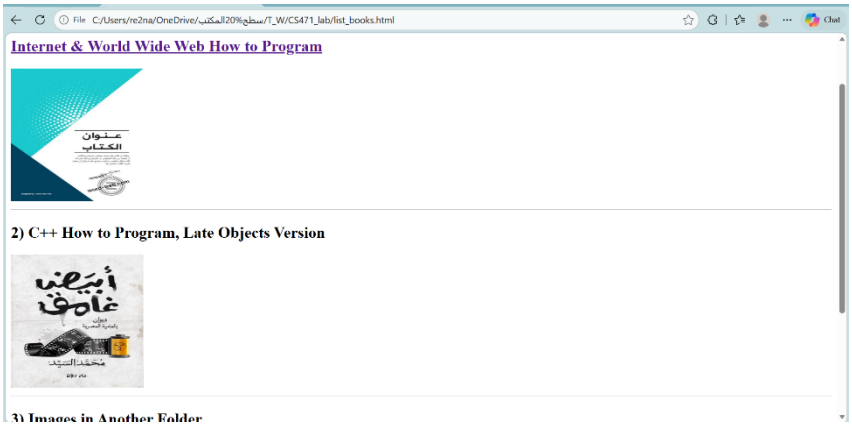
Step 3:

Name	Status	Date modified	Type	Size
qu-icon.png		٤٧/٠٩/١٥ ص ٠٥:٣٧	PNG File	38 KB
book1.jpg		٤٧/٠٩/١٥ ص ٠٥:٣٧	JPG File	31 KB
aboutus.html		٤٧/٠٩/١٥ ص ١٢:٢١	Microsoft Edge HTM...	1 KB
book2.jpg		٤٧/٠٩/١٥ ص ٠٥:٣٧	JPG File	131 KB
book3.jpg		٤٧/٠٩/١٥ ص ٠٥:٣٧	JPG File	138 KB
index.html		٤٧/٠٩/١٥ ص ١٢:٢٠	Microsoft Edge HTM...	1 KB
list_books.html		٤٧/٠٩/١٥ ص ١٢:٢٣	Microsoft Edge HTM...	1 KB
one_book.html		٤٧/٠٩/١٥ ص ١٢:٢٤	Microsoft Edge HTM...	1 KB

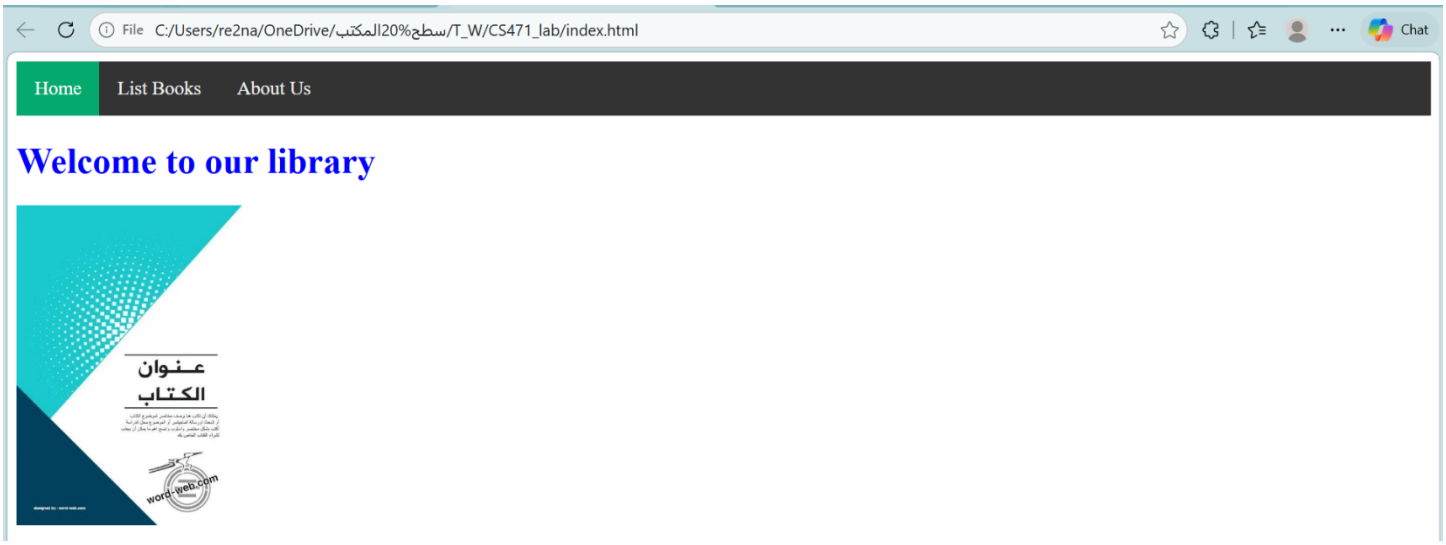
Step 4: To open an HTML5 examples from this task, open the folder where you saved the html files, and double click any file (index.html, for example), to open it in your default web browser.

Task 2: Style the previous HTML files with CSS in three ways (inline CSS, internal style element, and separated CSS file)

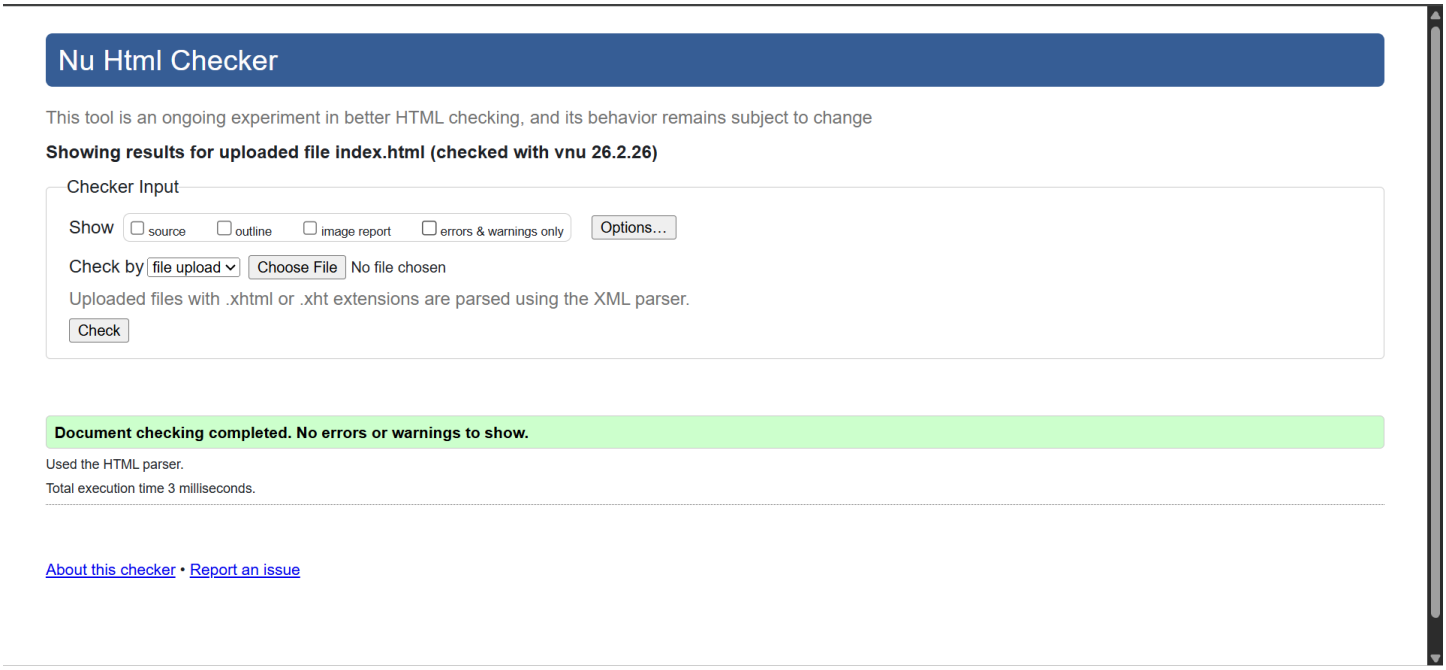
Step 1:



Step 4: Now double click on index.html, to open it in your default web browser.

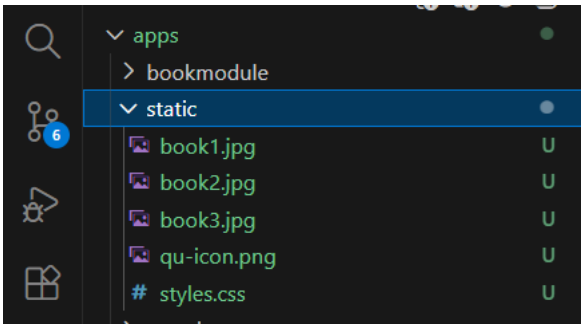


Task 3: W3C HTML5 Validation Service[1] to use proper HTML5 syntax to ensure that browsers process your documents properly.



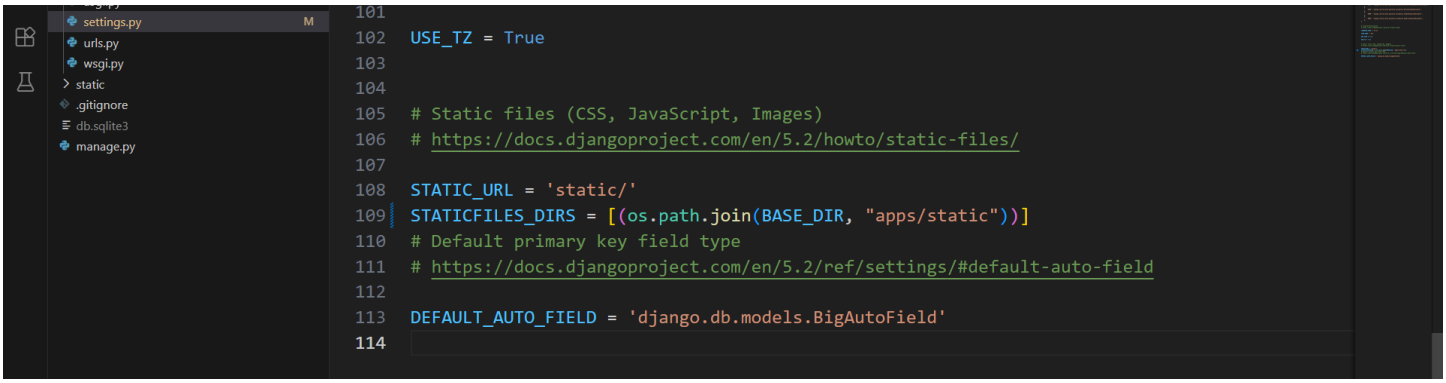
Task 4: Implementation and management of the previous HTML and static files (e.g. CSS and images) in Django framework.

Step 1: In the apps folder, create '`static`' folder



[1] The World Wide Web Consortium (W3C) provides a validation service (at validator.w3.org) for checking a document's syntax

Step 2:



Step 3: Move the style.css and four image files to '`apps/static`' folder

Step 4: From the root folder, navigate to 'apps/bookmodule/view.py', then, in views.py, add the following view functions:

```
def index(request):
    return render(request, "bookmodule/index.html")

def list_books(request):
    return render(request, 'bookmodule/list_books.html')

def viewbook(request, bookId):
    return render(request, 'bookmodule/one_book.html')

def aboutus(request):
    return render(request, 'bookmodule/aboutus.html')
```

Step 5: From the root folder, navigate to 'apps/bookmodule/urls.py', add a map to the view functions with `names` to the urlpatterns list:

```
urlpatterns = [
    path('', views.index, name= "books.index"),
    path('list_books/', views.list_books, name= "books.list_books"),
    path('<int:bookId>/', views.viewbook, name="books.view_one_book"),
    path('aboutus/', views.aboutus, name="books.aboutus"),
]
```

Step 6: Move HTML files to '`apps/templates/bookmodule`' folder, update URLs (for hyperlink, style, and images) with URL or Static Django template tag, this is an example for only index.html file, and you should do the same for the rest of html files:

```
<!DOCTYPE html>
<html>
{% load static %}<!-- load static -->
<head>
  <title>My First Index HTML Page</title>
  <meta name="keywords" content="Books, recent, Learning, library">
```

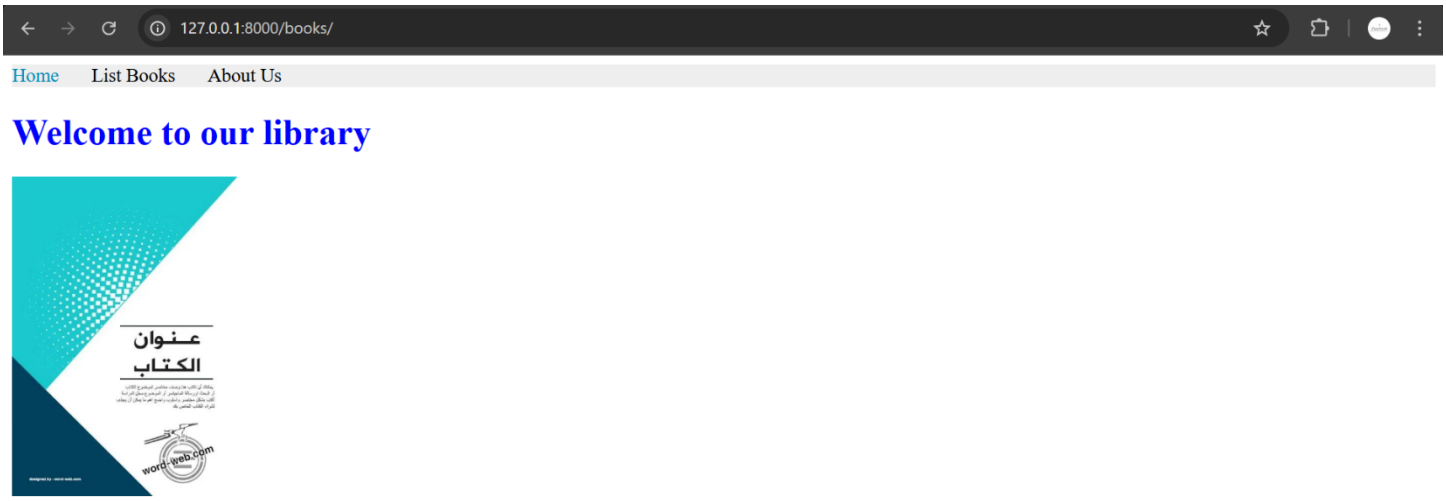
```
  <meta name="description" content="Books website">
  <meta name="author" content="author name">
  <link rel="icon" href="{% static 'qu-icon.png' %}"> <!-- static template tag -->
  <meta charset="UTF-8">

  <link rel="stylesheet" href="{% static 'styles.css' %}"> <!-- static template tag -
->
</head>
<body>

    <div class="topnav">
      <a class="active" href="{% url "books.index" %}">Home</a> <!-- URL template tag
-->
      <a href="{% url "books.list_books" %}">List Books</a> <!-- URL template tag -->
      <a href="{% url "books.aboutus" %}">About</a> <!-- URL template tag -->
    </div>
```

• • •

Step 7: Run the server if you haven't, open the web browser, and navigate to http://127.0.0.1:8000/books/.



Task 5: Additional implementation of Django tags to build one layout instead of repeating the layout for every HTML file.

Step 1: In '`apps/templates`' folder, add these new folders and files:

templates/

…

├── includes/

```
│   ├── header.html
│   ├── footer.html
└── layouts/
│   ├── base.html
```

Step 2: In `base.html`, add the following html content:

```
<!DOCTYPE html>
<html>
{% load static %}
<head>
  <title>{% block title %} {% endblock title %}</title>
  <meta name="keywords" content="Books, recent, Learning, library">
  <meta name="description" content="Books website">
  <meta name="author" content="author name">
  <link rel="icon" href="{% static 'qu-icon.png' %}">
  <meta charset="UTF-8">

  <link rel="stylesheet" href="{% static 'styles.css' %}"> <!-- include css file -->

  {% block stylesheets %}{% endblock stylesheets %}
</head>
<body>

    {% include "includes/header.html" %}

    {% block content %} {% endblock content %}

    {% include "includes/footer.html" %}

</body>
</html>
```

Step 3: In `header.html`, add the following html content:

```
{% with request.resolver_match.url_name as url_name %}
<div class="topnav">
    <a class="{% if url_name == 'books.index' %}active{% endif %}" href="{% url
"books.index" %}">Home</a>
    <a class="{% if url_name == 'books.list_books' %}active{% endif %}" href="{% url
"books.list_books" %}">List Books</a>
    <a class="{% if url_name == 'books.aboutus' %}active{% endif %}" href="{% url
"books.aboutus" %}">About</a>
</div>
{% endwith %}
```

If you want a simple html content for `header.html` without `class` attribute in a tag, then add this instead:

```
<div class="topnav">
    <a href="{% url "books.index" %}">Home</a>
```

```
    <a href="{% url "books.list_books" %}">List Books</a>
    <a href="{% url "books.aboutus" %}">About</a>
</div>
```

Step 4: update `index.html` to:

```
{% extends "layouts/base.html" %}

{% load static %}

{% block title %} My First Index HTML Page {% endblock title %}

{% block content %}
  <h1 style="color: blue;">This is heading 1 (Library)</h1> <!-- add inline css -->
  <p style="color: white;background-color: black;">This is a paragraph text </p> <!--
add inline css -->
  <a href="https://www.w3schools.com/html/default.asp">Click Here to learn more about
HTML & CSS</a>
  <hr>
  <!-- This is a comment in HTML, not render in the bowser -->
  <h2>This is heading 2</h2>
  <p>This is some other text. <br/> with newline <br/> and another line</p>
  <hr>

  <h2>This is heading 2</h2>
  <p>This is some other text.</p>
{% endblock content %}
```

Step 5: Update the remaining HTML files (aboutus.html, list_books.html, and one_book.html), similar to index.html, with the necessary changes according to Django template tags.

Step 6: Now from the browser navigate to http://127.0.0.1:8000/books/

Home List Books About

Welcome to our library